PROJECT REPORT

Project Title: Predicting Restaurant Inspection Risk
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Primary Paper:

> **Title:** Predicting Food Safety Compliance for Informed Food Outlet Inspections: A Machine Learning Approach
> **Authors:** Oldroyd, Rachel A.; Morris, Michelle A.; Birkin, Mark
> **Year:** 2021
> **Publication:** International Journal of Environmental Research and Public Health
> **Link:** https://pubmed.ncbi.nlm.nih.gov/34886362/

GitHub: https://github.com/katenip/ITCS-5154-ML-Project/tree/main

1. Introduction

1.1 Problem Statement

Restaurant health inspections are important because they protect the public from unsafe food practices, poor sanitation, and foodborne illness. They also encourage restaurants to follow health regulations and maintain safe conditions.

However, inspections require time, money, and staff resources. Local health departments are often responsible for many restaurants with limited personnel. Because of this, inspections are commonly based on fixed schedules, complaints, or broad risk categories rather than changing with current conditions.

This can create inefficiencies. Some lower-risk restaurants may receive the same attention as higher-risk locations, while some riskier restaurants may not be prioritized soon enough. Restaurants in the same category can also differ greatly in cleanliness, management, and past performance.

The goal of this project is to examine whether machine learning can help predict restaurant inspection risk using historical inspection data and local demographic information. If successful, this type of model could help health departments use resources more effectively and focus on higher-risk locations.

1.2 Motivation and Importance

I have long been interested in food systems and food safety, supported by coursework and previous experience working in an elementary school cafeteria. That experience helped show the importance of sanitation, safe food handling, and trust in food service environments.

I also have close relatives who are immunocompromised, which has highlighted how serious food safety can be for vulnerable populations. Foodborne illness can affect anyone, but it may create greater risks for children, older adults, and those with weakened immune systems.

Targeting higher-risk establishments earlier could improve public health outcomes by:

- Better directing limited health department resources
- Reducing exposure to unsafe food practices
- Encouraging more consistent compliance among establishments

1.3 Current Approaches and Limitations

Restaurant inspections are commonly based on routine schedules, complaint reports, and broad risk categories. While these methods provide structure, they may not fully reflect changing conditions at individual locations.

Restaurants in the same category can have very different risk levels based on management quality, staffing, cleanliness practices, and prior performance. Traditional systems may also make limited use of historical data or local environmental factors. Because of this, some higher-risk establishments may not be prioritized as effectively as possible.

1.4 Key Challenges

Several challenges exist when building a prediction model for restaurant inspection risk. Public inspection data may contain duplicates, missing values, or inconsistent restaurant names and addresses. Most restaurants also receive acceptable scores, creating class imbalance where risky outcomes are less common.

In addition, some important factors such as management quality or employee training are difficult to measure directly. These limitations make preprocessing and careful model design especially important.

1.5 Summary of Proposed Solution

The solution proposed in this project is that combining census data with factors such as restaurant type and historical inspection information can be used to build a prediction model that performs better than random guessing when identifying higher-risk locations. This project uses Mecklenburg County inspection records along with local census data to create features for machine learning analysis. The goal is to determine whether these combined factors can help predict restaurant inspection risk and support more informed inspection planning.

1.6 Contribution of This Project

This project applies ideas from prior research to a local county dataset rather than a larger national dataset. It combines restaurant inspection records with census-based demographic features and tests whether useful predictive patterns can still be found at the county level.

The project also examines threshold tradeoffs, feature importance, and practical use cases for inspection planning. This helps show how machine learning could be used as a decision-support tool in real public health settings.

2. Backgrounds

2.1 Primary Paper Overview

2.1.1 Study Objective

The primary paper used for this project was *Predicting Food Safety Compliance for Informed Food Outlet Inspections: A Machine Learning Approach* by Oldroyd, Morris, and Birkin (2021). The study examined whether machine learning could be used to predict food safety non-compliance in England and Wales using public data sources. The authors were motivated by limited inspection resources and delays in routine inspections, which created the need for more efficient inspection prioritization.

2.1.2 Dataset and Methods

The paper used a large dataset of more than 300,000 food outlets and combined inspection outcomes with neighborhood-level demographic, business type, and urbanness variables. Several machine learning models were tested, including linear Support Vector Machines, radial Support Vector Machines, and Random Forest classifiers. To address class imbalance, the study also used SMOTE oversampling and under-sampling methods.

2.1.3 Main Results

The best-performing model was a Random Forest trained with SMOTE data at a 1:1 ratio. This model achieved an AUC of approximately 0.87 and identified about 84% of non-compliant outlets in the testing set. The authors concluded that geographic and demographic context can provide useful signals when predicting inspection risk.

2.1.4 Strengths and Limitations

A major strength of the paper is its practical focus. Rather than replacing inspectors, the model was designed to support decision-making and help prioritize limited public health resources. Another strength is the use of openly available data, making the approach easier to reproduce. However, the study was based on England and Wales, meaning results may not directly transfer to other regions with different inspection systems or restaurant environments.

2.1.5 Relation to Current Project

This paper directly inspired the current project. My project follows a similar idea but applies it to Mecklenburg County restaurant inspection data in North Carolina. Instead of a national dataset, this project explores whether comparable predictive patterns can still be found using a smaller local dataset and U.S. Census features.

2.2 Related Study 1 – Foodborne Pathogen Prediction

2.2.1 Study Objective

A second paper reviewed for this project was *Machine Learning Prediction of Foodborne Disease Pathogens: Algorithm Development and Validation Study* by Wang et al. (2021). Unlike the first paper, which focused on predicting restaurant inspection compliance, this study focused on predicting which pathogen caused a foodborne illness case using surveillance data. The goal was to assist diagnosis when laboratory testing was unavailable or delayed.

2.2.2 Dataset and Methods

The researchers used data from China's National Foodborne Disease Surveillance Reporting System, which contained 2.6 million reported cases between 2011 and 2018. Only about 60,000 cases had confirmed pathogen testing, so the authors trained models using those labeled examples. They focused on four major pathogens: *Salmonella*, *Norovirus*, *Escherichia coli*, and *Vibrio parahaemolyticus*, which represented most confirmed cases.

Several machine learning models were tested, including Decision Trees, Random Forest, AdaBoost, and Gradient Boosted Decision Trees (GBDT).

2.2.3 Main Results

The best model was GBDT, which achieved approximately 69% classification accuracy across the four pathogen classes. While lower than the inspection-risk paper's AUC results, this was a more difficult multiclass medical prediction problem.

Important predictive features included time of illness, geographic latitude and longitude, patient age, diarrhea frequency, vomiting frequency, fever, and food name or type. Seasonal patterns were especially useful; for example, Norovirus appeared more often in autumn and winter, while Vibrio parahaemolyticus peaked in summer.

2.2.4 Strengths and Limitations

A strength of this paper is that it demonstrates how machine learning can extract useful patterns from public health surveillance systems, even when many cases are missing confirmed labels. It also highlights how nontraditional variables such as seasonality and geography can improve prediction. A limitation is that the dataset reflects healthcare reporting behavior and testing availability, meaning the labels may not perfectly represent all real-world infections.

2.2.5 Relation to Current Project

This paper is relevant to the current project because it reinforces a broader idea: machine learning can support food safety decisions by using indirect signals from structured data. In my restaurant inspection project, the target is inspection failure risk rather than pathogen type, but both studies rely on extracting predictive patterns from incomplete public health data.

2.3 Related Study 2 – Inspection Scores and Real Outcomes

2.3.1 Study Objective

A third paper reviewed for this project was As Clean as They Look? Food Hygiene Inspection Scores, Microbiological Contamination, and Foodborne Illness by Fleetwood et al. (2019). This study examined whether restaurant hygiene inspection scores are actually associated with measurable food safety outcomes. Unlike predictive modeling papers, this research focused on validating whether inspection ratings reflect real contamination risk and outbreak likelihood.

2.3.2 Dataset and Methods

The researchers used inspection and food sampling data from England, Wales, and Northern Ireland. They analyzed more than 37,000 microbiological food samples from the LAEMS database and an additional 7,115 samples from the UK Food Surveillance System (UKFSS). These samples were linked with official Food Hygiene Rating Scheme (FHRS) inspection scores ranging from 0 (urgent improvement necessary) to 5 (very good).

2.3.3 Main Results

Results showed that businesses with lower hygiene ratings were significantly more likely to have unsatisfactory microbiological samples. Premises that were not broadly compliant (ratings 0–2) consistently had higher contamination rates than those rated 3–5. In the UKFSS sample set, rating 5 establishments had a lower unsatisfactory sample rate (6.4%) compared with ratings 0–4 (8.2%).

The paper also examined foodborne illness outbreaks and found that broadly compliant establishments had fewer outbreaks per restaurant-year than noncompliant establishments. The authors estimated that if all businesses met broad compliance standards, foodborne illness cases linked to eating out could decrease by roughly 2.5% to 6.2%, while widespread noncompliance could substantially increase illness rates.

2.3.4 Strengths and Limitations

A major strength of this paper is that it connects inspection scores to real-world outcomes such as contamination and outbreaks rather than only administrative metrics. This supports the practical value of inspections as a public health tool. A limitation is that outbreaks are difficult to trace directly to specific restaurants, and inspections only capture conditions at one point in time.

2.3.5 Relation to Current Project

This paper is highly relevant to the current project because it supports the assumption that restaurant inspection scores contain meaningful signals about food safety risk. In my project, machine learning is used to predict future inspection outcomes. This study helps justify why predicting poor scores matters: lower inspection performance is associated with greater contamination risk and more foodborne illness.

2.4 Comparison to Main Method

2.4.1 Similarities to the Current Project

The three research papers reviewed in this section are related to the main method used in this project, but each approaches food safety from a different angle. The primary paper by Oldroyd et al. most directly matches this project because it used machine learning to predict inspection non-compliance using demographic and business features. My project follows a similar idea but applies it to Mecklenburg County restaurant data using U.S. Census variables and a smaller local dataset.

2.4.2 Differences Across the Studies

The second paper by Wang et al. focused on predicting foodborne disease pathogens rather than inspection outcomes. Although the target variable is different, the study still demonstrates how machine learning can identify useful patterns in public health data and support decision-making when direct information is limited.

The third paper by Fleetwood et al. did not build a prediction model, but it is important because it showed that lower inspection scores are associated with greater microbiological contamination and more foodborne illness risk. This helps support the value of predicting inspection outcomes, since poor scores are linked with meaningful public health consequences.

2.4.3 Contribution of the Current Project

Together, these papers support the overall approach of this project. They show that structured public data, demographic information, and inspection records can be used to better understand food safety risk. My project extends these ideas by testing whether similar methods can be applied successfully at the county level using publicly available U.S. data.

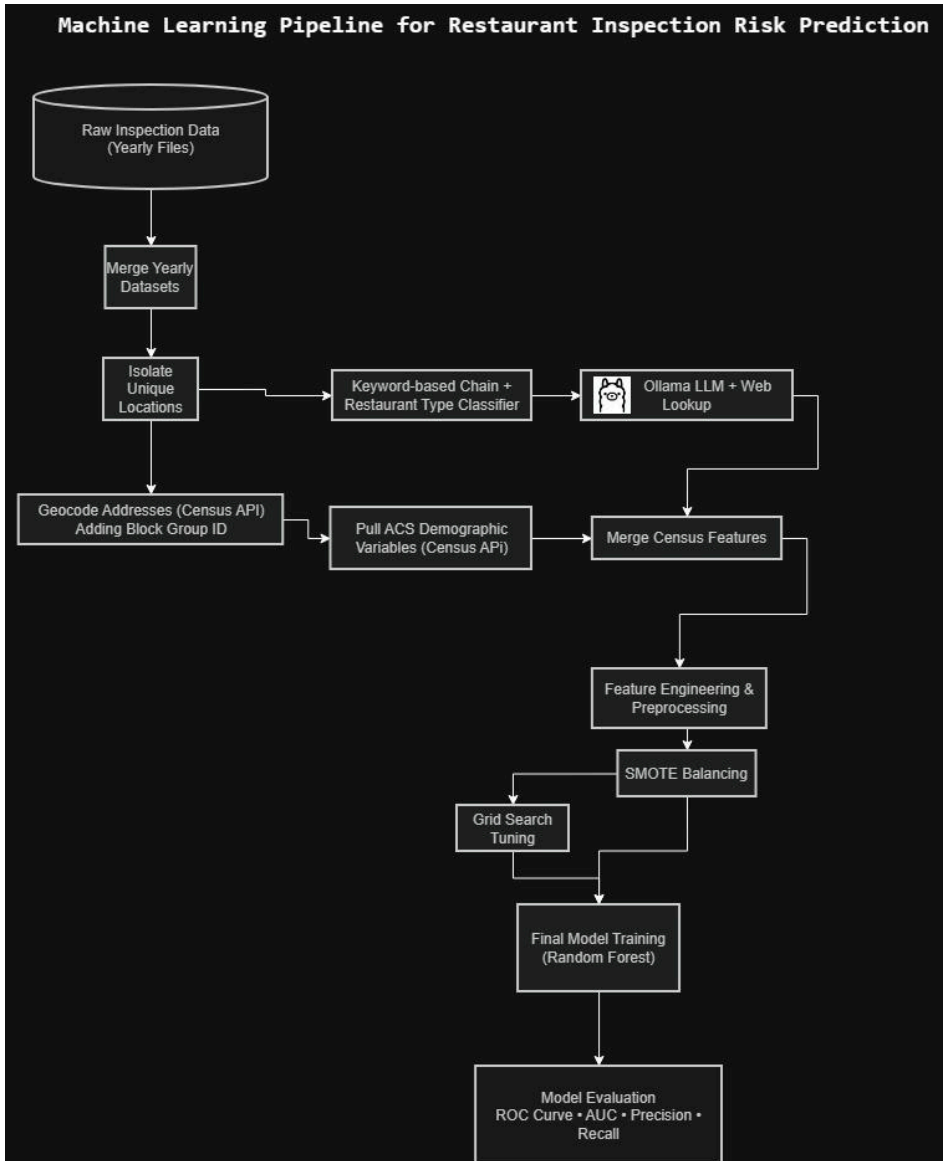
3. Methods

3.1 Overall Framework

3.1.1 Pipeline Steps

- Collect multiple years of restaurant health inspection data using 2015–2024 data from Mecklenburg County.
- Merge yearly datasets into one master dataset
- Isolate unique restaurants / locations
- Geocode restaurant addresses using U.S. Census Geocoder API adding census features like Block Group
- Pull ACS Census demographic and housing variables for each block group
- Merge Census features with inspection dataset
- Apply keyword-based classifier to detect restaurant chains and business types
- Use Ollama LLM (llama 3.2) + web lookup/manual review to classify remaining unknown restaurants
- Final feature engineering and preprocessing
- Apply SMOTE to balance minority risk class
- Train Random Forest model
- Perform Grid Search hyperparameter tuning
- Evaluate model using ROC curve, confusion matrix, precision, recall, AUC, feature importance

3.1.2 Diagram



3.2 Data Sources

Two primary data sources were used in this project: restaurant inspection records and U.S. Census demographic data.

3.2.1 Restaurant Inspection Data

The restaurant inspection data was obtained from the Mecklenburg County Environmental Health public database. These records included information such as establishment name, address, inspection date, and inspection score. Multiple years of data were collected in order to analyze historical inspection outcomes and build a larger training dataset.

3.2.2 Census Demographic Data

The second source was demographic information from the U.S. Census Bureau American Community Survey (ACS). Census variables were used to describe neighborhood characteristics surrounding each restaurant location. Examples included median household income, unemployment levels, renter occupancy, vehicle access, racial and ethnic composition, and population measures.

The two datasets were combined by geocoding restaurant addresses and linking each location to its corresponding census geography. This created a richer dataset that included both business-level inspection information and surrounding community context for machine learning analysis.

3.3 Data Preprocessing

Several preprocessing steps were required before model training. Because the project combined inspection records from multiple years with external demographic data, cleaning and standardization were necessary to create a usable dataset.

3.3.1 Data Integration and Standardization

Inspection records from multiple years were collected and merged into a single dataset. Column names, formats, and variable types were standardized so records from different years could be analyzed consistently.

3.3.2 Duplicate Removal and Cleaning

The raw data contained duplicate entries, repeated inspections, and inconsistent restaurant names or addresses. These issues were addressed through cleaning and deduplication procedures. Similar business names were standardized when possible, and incomplete or clearly invalid records were removed.

3.3.3 Geocoding and Census Matching

Restaurant addresses were geocoded in order to assign census geographic identifiers such as block groups or tracts. This made it possible to merge demographic variables from external census datasets. Records that could not be matched reliably were reviewed or excluded when necessary.

3.3.4 Missing Values and Target Labeling

Missing values were handled using reasonable defaults or row removal depending on the variable and amount of missing data. After cleaning, a target label was created using selected inspection score thresholds to classify establishments into lower-risk and higher-risk outcomes.

3.3.5 Local LLM-Assisted Business Type Classification

After applying keyword-based chain and category rules, 1,105 unique locations remained uncategorized. These establishments were passed to a local llama3.2 instance running via Ollama, with each prompt including the establishment name, address, and city along with instructions to assign exactly one category from the 23 allowed types. The model returned a category, confidence score, and short reasoning string for each prediction. Because no ground truth labels existed for these restaurants, the Ollama-assigned categories were not formally validated and some misclassifications are likely. This step was used only as a fallback after deterministic rules were exhausted, so any labeling errors are limited to this subset of establishments rather than the full dataset.

3.4 Feature Engineering

Feature engineering was used to convert raw inspection and census data into variables that could be used for machine learning. The goal was to create features that captured business characteristics, historical patterns, and neighborhood context that may be associated with restaurant inspection risk.

3.4.1 Business Characteristics

Business-related features included restaurant category labels such as fast food, full service, grocery deli, bakery, and other establishment types. Chain indicators were also considered because larger chains may differ from independent restaurants in staffing, procedures, and operational consistency.

3.4.2 Historical Inspection Features

Historical inspection features were created from previous inspection records when available. These included prior scores, rolling averages, inspection counts, and patterns that may reflect past compliance history. Locations with repeated issues may present different future risk than consistently high-performing establishments.

3.4.3 Neighborhood Demographic Features

Neighborhood demographic features were added using U.S. Census American Community Survey data linked through geocoded restaurant addresses. These variables included median income, unemployment levels, renter occupancy, vehicle access, racial and ethnic composition, and local population measures.

3.4.4 Encoding and Model Preparation

Categorical variables were converted into numerical form so they could be used by the model. This allowed restaurant categories and other non-numeric variables to be incorporated into the training process.

3.5 Algorithm Selection

Several machine learning methods were considered for this project, with the final model primarily focusing on a Random Forest classifier. The selected algorithm needed to perform well on structured public data while remaining practical to train and interpret.

3.5.1 Performance on Tabular Data

Random Forest was selected because it performs well on structured tabular datasets, can model nonlinear relationships, and captures interactions between variables without requiring extensive manual tuning.

3.5.2 Robustness to Real-World Data

Another advantage of Random Forest is that it is relatively robust to noisy data and less sensitive to scaling requirements than some other algorithms. This was useful because the dataset combined multiple feature types, including demographic values, business categories, and historical inspection information.

3.5.3 Interpretability

Random Forest also provides feature importance measures, which helped identify which variables were most useful in predicting restaurant inspection risk. This supported one of the project goals: not only making predictions, but also understanding what factors may be associated with higher-risk outcomes.

3.5.4 Alternative Models Considered

Additional models such as Support Vector Machines and boosting methods were considered based on the primary research paper. However, Random Forest offered a strong balance between predictive performance, interpretability, and practical implementation for this dataset.

3.6 Class Imbalance Handling

Class imbalance was an important issue in this project because most restaurants received acceptable inspection scores, while lower-scoring or higher-risk outcomes were less common. This created an uneven target distribution where the model could become biased toward predicting the majority class.

3.6.1 Imbalanced Target Distribution

Because most inspections resulted in acceptable scores, the number of higher-risk cases was much smaller than the number of lower-risk cases. Without correction, the model could favor majority-class predictions while failing to learn minority-class patterns.

3.6.2 Use of SMOTE

To address this problem, Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. SMOTE creates synthetic examples of the minority class rather than simply duplicating existing records. This helps produce a more balanced training set and improves the model's ability to recognize higher-risk establishments.

3.6.3 Preventing Data Leakage

SMOTE was applied only to the training set and not to validation or testing data. This was done to prevent data leakage and ensure that evaluation results reflected true model performance on unseen records.

3.7 Model Training and Validation

The dataset was divided into training and testing groups so that model performance could be evaluated on unseen data. Older inspection records were used for training, while newer records were reserved for testing in order to better reflect how the model would be used in practice to predict future inspection outcomes.

3.7.1 Train-Test Split Strategy

The data was separated into training and testing sets using a time-based approach. Older inspection records were used for model training, while newer records were held out for final testing. This better simulated real-world forecasting conditions.

3.7.2 Model Training

The Random Forest classifier was trained using the selected engineered features after preprocessing and class balancing steps were completed. Hyperparameters such as the number of trees, tree depth, and class weighting were adjusted through experimentation to improve performance.

3.7.3 Validation and Generalization

After training, the model was evaluated on the held-out test set rather than only the training data. This helped measure how well the model generalized to new restaurant records and reduced the risk of overestimating performance through overfitting.

3.7.4 Final Model Selection

The final model was selected based on its balance between predictive accuracy and practical usefulness for identifying higher-risk establishments.

3.8 Evaluation Metrics

Several evaluation metrics were used to measure model performance. Because this project focused on identifying higher-risk restaurants rather than only maximizing overall accuracy, multiple metrics were considered.

3.8.1 Primary Metric: ROC-AUC

The primary metric used was Receiver Operating Characteristic Area Under the Curve (ROC-AUC). ROC-AUC measures how well the model can distinguish between higher-risk and lower-risk establishments across many decision thresholds. This metric was especially useful because it evaluates ranking ability rather than relying on one fixed cutoff value.

3.8.2 Precision and Recall

Additional metrics included precision and recall. Precision measures how often restaurants predicted as high-risk were actually high-risk, while recall measures how many true high-risk restaurants were successfully identified. These metrics help evaluate the tradeoff between false alarms and missed risky establishments.

3.8.3 Confusion Matrix Analysis

The confusion matrix was used to examine false positives, false negatives, true positives, and true negatives. This provided a clearer picture of model behavior beyond a single summary score.

3.8.4 Threshold Analysis

Threshold analysis was also performed because changing the classification cutoff can improve recall or precision depending on operational goals. In a real inspection setting, decision makers may prefer a higher recall model to reduce the chance of missing risky establishments.

4. Experiments

4.1 Experimental Setup

The experiments were designed to test whether restaurant inspection risk could be predicted using historical inspection data, business type information, and local demographic features. After preprocessing, the dataset was divided into training and testing sets so that performance could be measured on unseen records.

4.1.1 Train-Test Strategy

Older inspection records were primarily used for training, while newer records were reserved for testing. This better simulated real-world deployment where past data is used to predict future inspection outcomes.

4.1.2 Model Configuration

A Random Forest classifier was selected as the primary model because it performs well on structured tabular data and can capture nonlinear relationships between variables. SMOTE was applied only to the training data to reduce class imbalance. Multiple score thresholds were also tested when defining higher-risk outcomes in order to evaluate how target definitions affected model behavior.

4.1.3 Evaluation Criteria

Several evaluation metrics were used, including ROC-AUC, precision, recall, and confusion matrix analysis. These metrics allowed both predictive quality and practical tradeoffs to be examined.

4.2 Main Results

The final model showed meaningful predictive ability and performed better than random guessing. The strongest overall model achieved an ROC-AUC of approximately 0.671, indicating moderate ability to rank higher-risk restaurants above lower-risk restaurants.

4.2.1 Interpretation of Performance

Although this performance was below the results reported in the primary paper, it still demonstrated that inspection records and census-based contextual data contain useful predictive signals. The model identified patterns beyond simple chance and suggests that restaurant risk is partially predictable using public data.

4.2.2 Practical Value

In an operational setting, even moderate ranking ability may help inspection agencies focus limited resources on establishments with elevated predicted risk rather than relying only on fixed schedules or random selection.

4.3 Reproducing the Paper

The primary research paper reported stronger performance, with an AUC near 0.87 using a large national dataset from England and Wales. In comparison, this project achieved lower performance on Mecklenburg County data.

4.3.1 Reasons for Performance Differences

Several factors likely explain this difference. The original study used a much larger dataset with more than 300,000 records, while this project relied on a smaller county-level dataset. The paper also included variables such as urbanness and additional geographic classifications that were not directly available here. In addition, local public inspection data may contain more noise, duplicates, and inconsistencies than a curated research dataset.

4.3.2 Reproduction Results

Even with these differences, this project applied the same approach to a different, smaller dataset and validated the general direction.

4.4 Additional Experiments

Additional experiments were conducted by testing multiple score thresholds for defining higher-risk restaurants. Thresholds such as 95, 92, and 90 were explored.

4.4.1 Threshold Effects

Higher thresholds increased the number of restaurants labeled as risky, while lower thresholds created a stricter definition of poor performance.

4.4.2 Impact on Model Behavior

These experiments showed that target definition strongly affects class balance and model behavior. A stricter threshold often reduced the number of positive cases, making prediction more difficult. A broader threshold increased recall opportunities but could reduce precision.

4.4.3 Operational Implications

This demonstrates that model design should consider operational goals rather than relying on one arbitrary score cutoff.

4.5 Feature Importance Observations

Feature importance analysis was used to better understand which variables contributed most to model predictions.

4.5.1 Historical Features

Historical inspection performance was one of the most useful predictors when available. Prior scores and compliance trends likely contain meaningful information about future outcomes.

4.5.2 Business Type Variables

Restaurant category variables also contributed useful information, indicating that some establishment types may have different average risk patterns.

4.5.3 Demographic Context

Neighborhood demographic variables added additional predictive value, supporting the idea that surrounding context may influence restaurant outcomes.

4.5.4 Multi-Factor Nature of Risk

No single feature fully explained results, suggesting that inspection risk is driven by multiple interacting factors.

4.6 Model Strengths and Weaknesses

4.6.1 Strengths

One strength of the model is that it uses publicly available data and can be reproduced at relatively low cost. The Random Forest approach also handles mixed feature types well and provides interpretable importance measures.

4.6.2 Weaknesses

Predictive performance was moderate rather than high, meaning it should not be used as a fully automated decision system. The model was also limited by data quality, unavailable hidden variables, and the smaller size of the county dataset.

4.7 Personal Thoughts / Discussion

This project showed that machine learning can provide useful support in public health resource allocation even when predictions are imperfect.

4.7.1 Decision Support Rather Than Replacement

The model should be viewed as a decision-support tool rather than a replacement for inspectors. Human expertise, follow-up investigations, and policy judgment would still be necessary.

4.7.2 Broader Takeaway

Overall, the results suggest that public data can be used in meaningful ways to improve how inspections are prioritized and planned.

5. Conclusions

5.1 Project Summary

This project examined whether machine learning could be used to predict restaurant inspection risk using public inspection records and local demographic data. Mecklenburg County inspection data was combined with U.S. Census information to create a feature-rich dataset for analysis. A Random Forest classification model was then trained to estimate the likelihood that a restaurant would receive a lower inspection outcome in a future inspection.

5.1.1 Overall Workflow

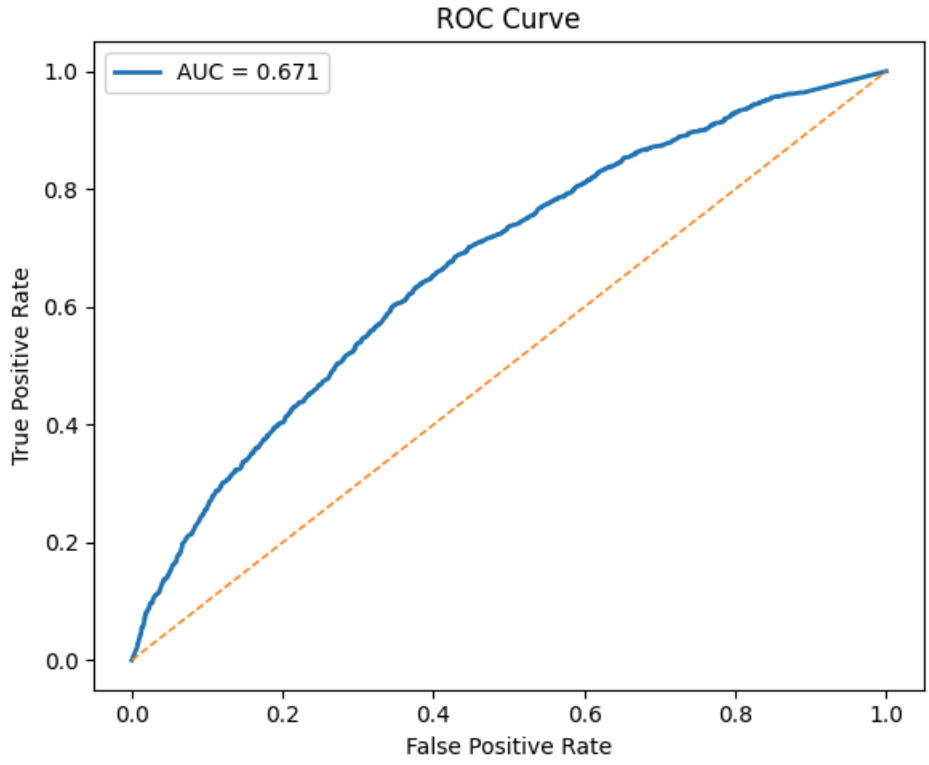
The project involved collecting multi-year inspection records, preprocessing and cleaning the data, geocoding restaurant locations, merging census features, engineering predictive variables, handling class imbalance with SMOTE, and evaluating a supervised learning model.

5.2 Main Findings

The final model demonstrated meaningful predictive ability and performed better than random guessing when identifying higher-risk establishments. The strongest model achieved an ROC-AUC of approximately 0.671, indicating moderate ability to rank risk across restaurants.

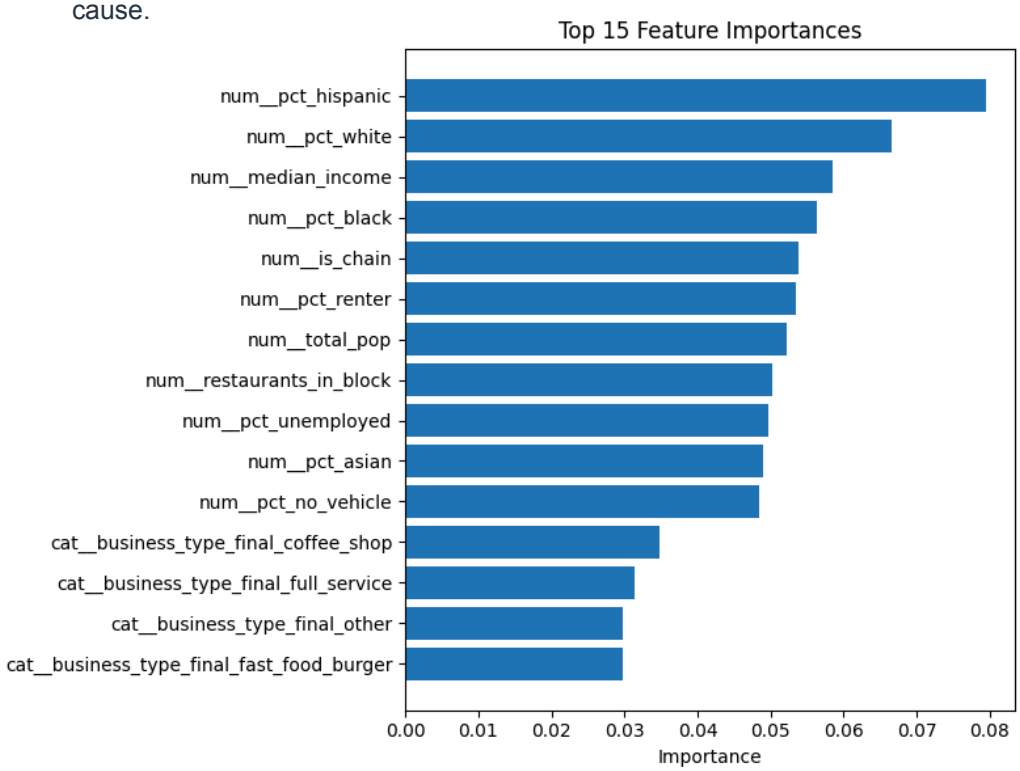
5.2.1 Predictive Performance

Although the model did not reach the performance level of the primary research paper, it still showed that useful signals exist within public inspection and demographic data. This suggests that county-level data can support practical risk ranking.



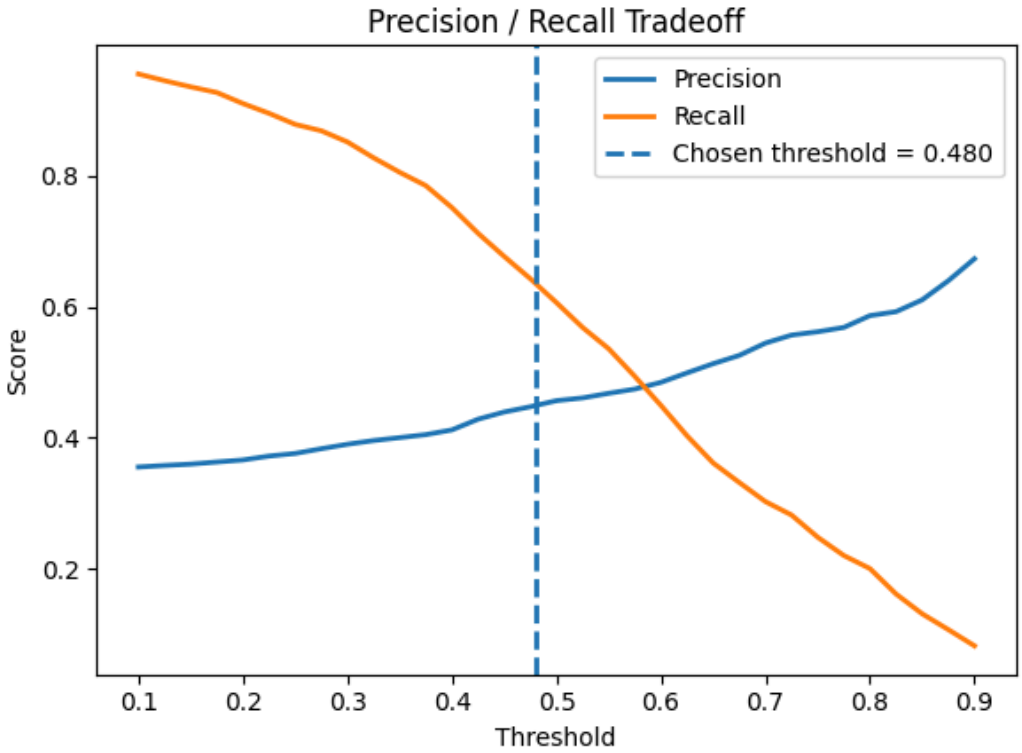
5.2.2 Important Features

Feature importance analysis suggested that restaurant chain status, business category, historical inspection patterns, and neighborhood demographic variables all contributed useful predictive information. This indicates that inspection outcomes are influenced by multiple interacting factors rather than one single cause.



5.2.3 Threshold Tradeoffs

The project also showed that threshold selection creates an important tradeoff between precision and recall. Lower thresholds captured more risky establishments, while higher thresholds reduced false positives.



5.3 Overall Thoughts

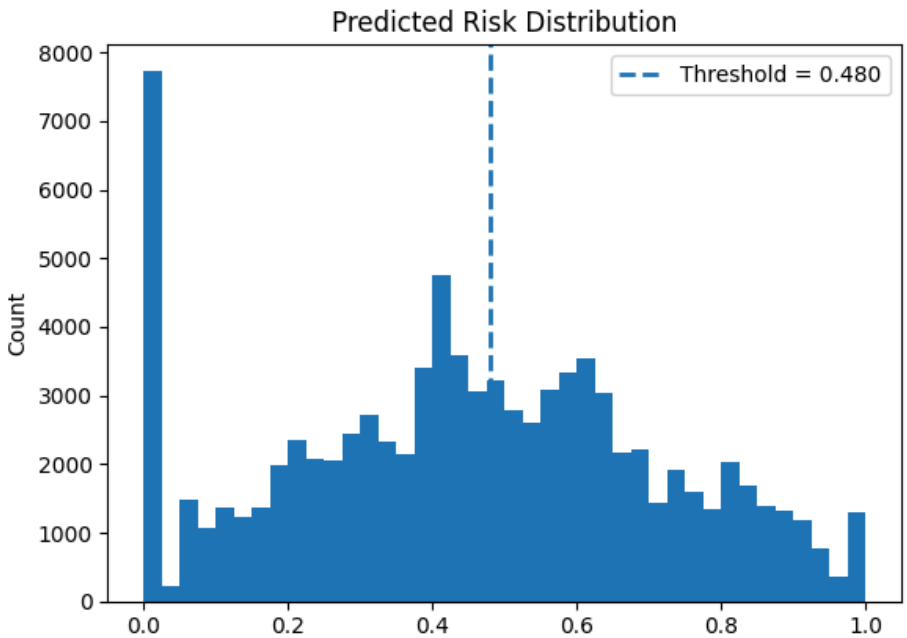
Although the model is not accurate enough to replace inspectors, it could still be valuable as a decision-support tool. In many real-world settings, a model does not need perfect accuracy to be useful. If limited inspection resources can be directed more efficiently than random scheduling or broad category rules, public health outcomes may improve.

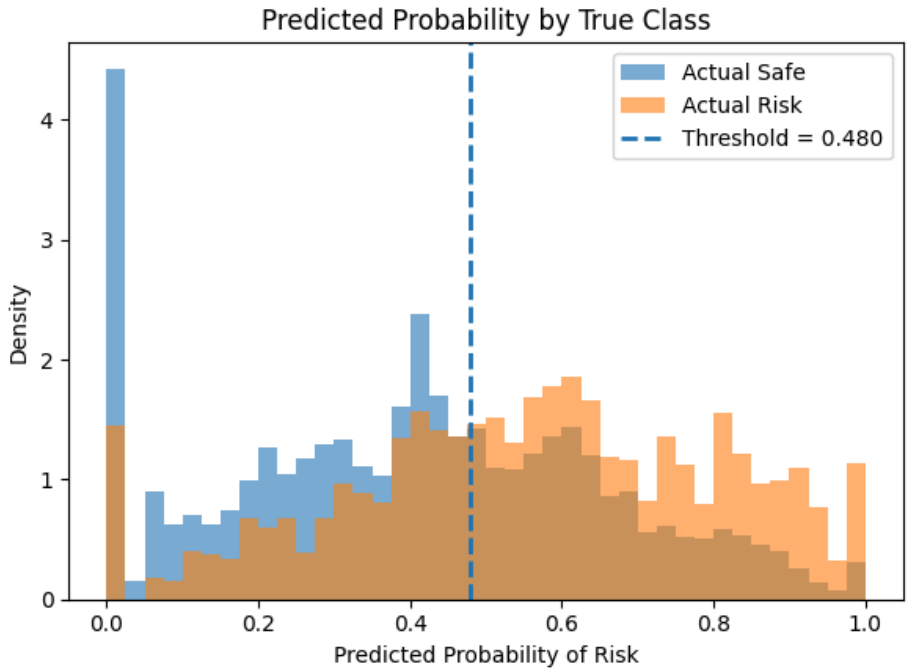
5.3.1 Practical Use Case

The strongest value of this model would likely be prioritization rather than automation. Inspectors could use predictions as one factor when deciding where to focus limited time and staffing resources.

5.3.2 Broader Significance

This project also demonstrated how public data can be transformed into actionable insights through machine learning. Even with imperfect data, meaningful patterns were still identified.





5.4 Limitations

This project has several limitations. The dataset was limited to a single county and was smaller than the national dataset used in the primary research paper.

5.4.1 Data Quality Constraints

Public inspection data may contain missing values, inconsistent records, duplicate entries, or noisy labels. These issues can reduce model performance.

5.4.2 Hidden Variables

Important factors such as management quality, staffing practices, employee training, and day-to-day cleanliness are difficult to measure directly using public records.

5.4.3 Generalizability

The model was trained on Mecklenburg County data and may not transfer directly to other counties or states without retraining.

5.5 Future Work

Future work could improve this project by expanding both data sources and modeling methods.

5.5.1 Expanded Datasets

Using larger multi-county or statewide datasets could improve model robustness and generalizability.

5.5.2 Additional Features

Future versions could include detailed violation text, review data, business turnover rates, or more granular geographic variables.

5.5.3 Advanced Models

Additional machine learning methods such as XGBoost, LightGBM, or neural networks could be tested. Time-series approaches may also better capture changing restaurant risk over time.

5.5.4 Explainable AI

Future systems could provide interpretable alerts explaining why a restaurant was flagged rather than only outputting a score.

5.6 Sharing Agreement

I do not agree to share this work as an example.

6. References

6.1 Studies

6.1.1 Primary Paper

Predicting Food Safety Compliance for Informed Food Outlet Inspections: A Machine Learning Approach
Oldroyd, Rachel A.; Morris, Michelle A.; Birkin, Mark
2021 — International Journal of Environmental Research and Public Health
Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC8656817/

6.1.2 Secondary Paper

Wang H, Cui W, Guo Y, Du Y, Zhou Y. Machine Learning Prediction of Foodborne Disease Pathogens: Algorithm Development and Validation Study. JMIR Med Inform. 2021 Jan 26;9(1):e24924. doi: 10.2196/24924. PMID: 33496675; PMCID: PMC7872834.
Link: https://pubmed.ncbi.nlm.nih.gov/33496675/

6.1.3 Tertiary Paper

Janet Fleetwood, Shamim Rahman, Darren Holland, David Millson, Laura Thomson, Guy Poppy, As clean as they look? Food hygiene inspection scores, microbiological contamination, and foodborne illness, Food Control, Volume 96, 2019, Pages 76-86, ISSN 0956-7135
Link: https://doi.org/10.1016/j.foodcont.2018.08.034

6.2 Data:

6.2.1 Inspection Data

Mecklenburg County Environmental Health. (2015 - 2024). Food Establishment Inspection Data (Mecklenburg County, NC). Retrieved from https://public.cdpehs.com/NCENVPBL/ESTABLISHMENT/ShowESTABLISHMENTTablePage.aspx?ESTTST_CTY=60

6.2.2 Census Data

U.S. Census Bureau. American Community Survey 1-Year
Estimates (2023). https://data.census.gov/